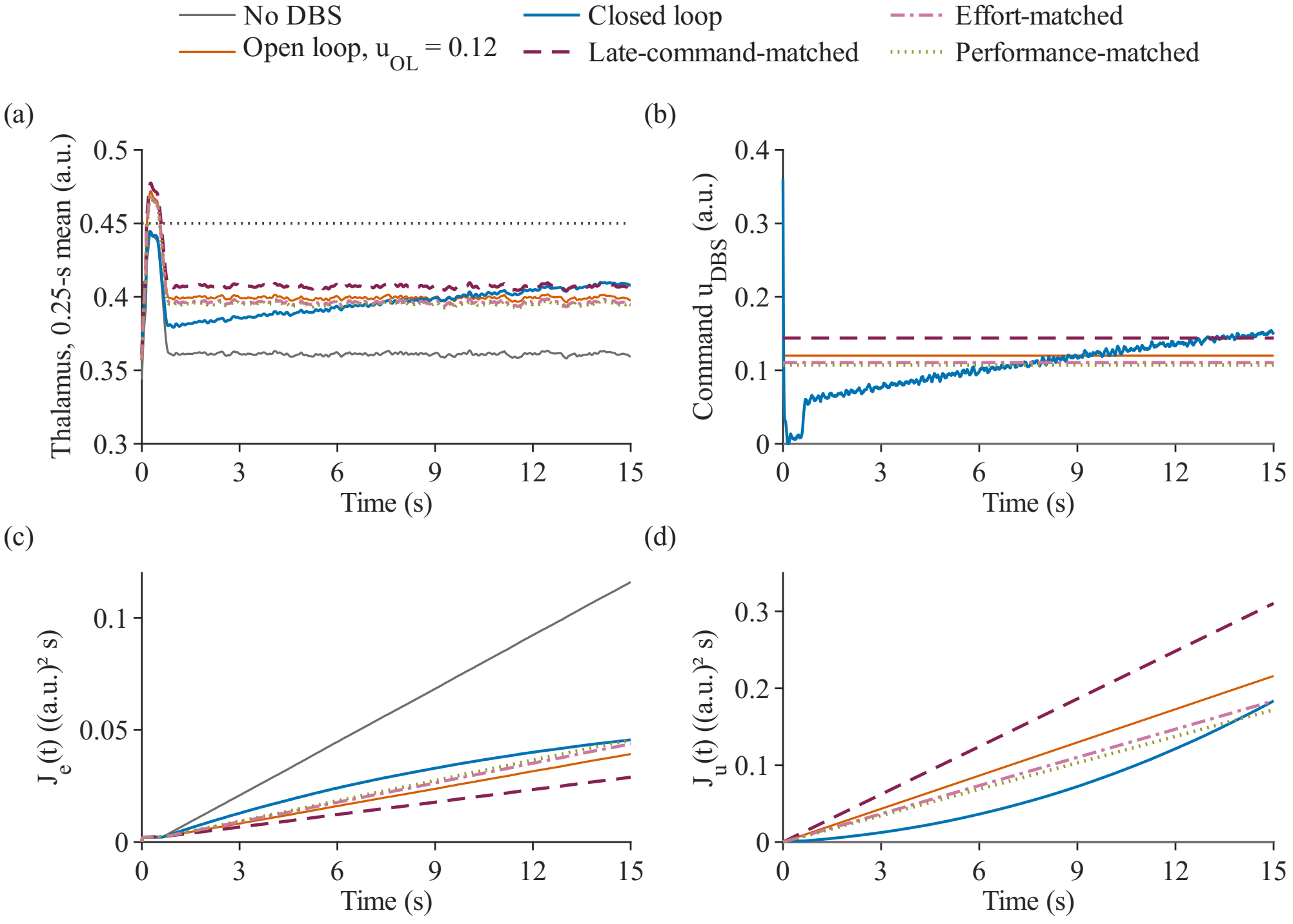

No DBS
Open loop, u_OL = 0.12
Closed loop
Late-command-matched
Effort-matched
Performance-matched
(a)
Thalamus, 0.25-s mean (a.u.)
0.5
0.45
0.4
0.35
0.3
0
3
6
9
12
15
Time (s)
(b)
Command u_DBS (a.u.)
0.4
0.3
0.2
0.1
0
0
3
6
9
12
15
Time (s)
(c)
J_e(t) ((a.u.)^2 s)
0.1
0.05
0
0
3
6
9
12
15
Time (s)
(d)
J_u(t) ((a.u.)^2 s)
0.3
0.2
0.1
0
0
3
6
9
12
15
Time (s)